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UAV Autonomous Landing Technology Based on AprilTags Vision Positioning Algorithm

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Abstract: The issue of the autonomous landing of the quadrotor unmanned aerial vehicle (UAV) has become a hot topic in recent years. How to improve the positioning accuracy of autonomous landing, reduce the calculation time, and improve the dynamic tracking effect are all problems that need to be solved. In this paper, an onboard solution to the tracking and landing problems of UAV and ground vehicle is presented. The AprilTags visual positioning algorithm based on monocular camera is used to locate and estimate the UAV's states. The platform used for UAV landing is labeled with different size of tags so that the drone can get good positioning accuracy at different heights. Combined with the PID control algorithm, the performance of dynamic tracking is improved. The experimental results show that the proposed method has good positioning accuracy and fast calculation speed.

Key Words: Quadrotor, autonomous landing, AprilTags, PID

1 Introduction

In recent years, the quadrotor UAV has become a hot spot in the field of drone with its simple mechanical structure and unique flight mode. However, without remote operation, the UAV cannot perform the flight mission autonomously. It means that the remote operation is the main way of working; the drone takes off and land by manual operation. It is because that the problem of autonomous location has not been solved well. With the future demand for UAV in military and civil fields, such as reconnaissance, search, rescue, and defense, there is an urgent need for fully autonomous flight UAVs that do not rely on manual control and satellite navigation.

At present, the widely used positioning methods are mainly based on Global Positioning System (GPS). However, in some special environment, for example, between buildings, inside large buildings, GPS signals are weak, and traditional positioning methods are hard to work [1]. Since the positioning method based on computer vision has the advantages of small size, light weight, low price and high precision of the airborne camera, the visual navigation gradually becomes the main trend of the research on the autonomous landing of the drone.

The autonomous landing of the UAV is the most critical event because crash happens frequently during its autonomous landing [2]. Generally, the landing operation requires accurate position information and control algorithm, which can be provided by the onboard navigation equipment and flight control system [3]. Once combined with computer vision algorithms, the positioning and navigation problems of the UAV can be effectively solved when GPS is weak, even missed. The research results of UAV combined with vision are widely used in video surveillance, intelligent transportation, post-disaster search and rescue, human-computer interaction, military applications and other fields [4].

The autonomous landing of drones mainly involves the identification and tracking of the landing platform and the motion control of drones. The research on the autonomous landing technology of the drone includes two parts: ground station processing and onboard processing [5]. The ground station communicates with the drone through the wireless local area network and receives the image information sent back by the drone by means of the wireless picture transmission device. After the processing is completed, it is sent back to the drone control system. In [6], a fixed-point landing system for quadrotor UAV has been designed, which is based on OpenCV, and a visual landing system for transmitting data through WIFI is implemented in the laboratory. [7] uses dm3730, a ground station processor with better performance, to process the image data collected by the UAV, but there is still a large time delay and distortion in the whole process. The processing of each image frame is required more than 1s.

The above research cannot guarantee the real-time and accuracy of unmanned positioning. The UAV is equipped with a variety of onboard processors to process the image in real time today. With the help of an onboard processor, [8] proposes a UAV landing algorithm based on global search, which solves the problem of how to search for ground landing platforms at high altitude with significant visual. Both the ground and the bottom of the drone are equipped with visual tags for bidirectional search, which reduces the error rate of the search, but there is no study about autonomous landing at a height close to the ground. The recognition problem of ground landing targets belongs to an important direction in the semantic classification of image scenes. [9] uses the Histogram of Oriented Gradient (HOG) algorithm to extract scene features and propose a field scene recognition method for the drone by combining the convolutional neural network. The UAV can land automatically according to the identified scene， but there is some mistakenly identified of the algorithm, which may lead to the failure of the landing. Harris algorithm is used to

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extract feature points from preprocessed images by using binocular camera [10], and NCC algorithm is used for feature matching to realize autonomous navigation of UAV, but the binocular camera requires complex calibration work. In addition, the calculation algorithm of the matching algorithm needs too much computation. [11] uses the monocular camera to realize the pose estimation of the drone which aims at the environment with beacons, which is more binocular. The camera has faster calculation speed and anti-interference ability, but the experimental scene has particularity.

In view of the research work and content above, this paper studies the autonomous landing and tracking for UAV and proposes an onboard solution for the drone. Some issues including motion control, target detection and system communication are also addressed in this study. The Apriltags localization algorithm based on monocular vision is used to locate and estimate the attitude of UAV. By setting different sizes of the tags as the landing target, the UAV can locate accurately at different altitudes. Several experiments show that the method proposed in this paper has good landing performance and tracking accuracy.

2 AprilTags visual positioning algorithm

Considering the performance of the onboard processor, this paper selects the AprilTags visual positioning algorithm based on monocular vision. AprilTags is a visual reference library that is widely used in AR, robotics, and camera calibration. With a special pattern, the markers can be quickly detected, and the relative position can be calculated.

2.1 AprilTags positioning algorithm principle

Apriltags system consists of two main parts: labeling detector and coding system. The detector is used to find the relative position and angle between the AprilTags tag and the camera. The coding system is used to extract the information contained in AprilTags. The AprilTags positioning algorithm has two steps: tag detection and attitude estimation. The detection flow chart of the algorithm is shown in Fig. 1.

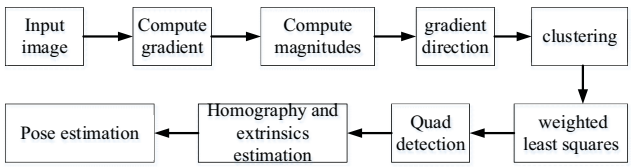


Fig. 1: Algorithm flowchart

The first step of AprilTags detection is to detect the line segments in the image. Firstly, the gradient direction and amplitude of each pixel are calculated. Using the graph-based method [11], pixels with similar gradient directions and amplitudes are clustered together. The lines in the image are then detected The second step is to detect the quadrilateral. The detection process of the quadrilateral is a recursive depth priority search with a depth of 4. The first layer, in turn, takes each line segment as the actual line segment. From the second layer to the fourth layer, to find the segment which is close enough to the end of the starting line, to increase the robustness of the most closed region and the segmentation error by adjusting the threshold value, and to use a two-dimensional lookup table to speed up the search for the segment closest to a certain point thus the correct quadrilateral can be found. After finding the lines and squares in the image and comparing them with the tags in the standard library, we can determine whether they are the tags we want to recognize. The third step is to calculate the relative position and attitude between the tag and the camera. After giving the calibrated camera inner parameter matrix and the size of the detected label, the homotropic matrix is calculated according to the Direct Linear Transform (DLT) algorithm. Then the relative position and attitude between the tag and the camera can be solved, which is discussed in detail in [12].

2.2 The design of the landing target

Since the UAV need to be able to detect landing targets at different altitudes, two labels of different sizes have been designed as landing targets. As shown in Fig.2.

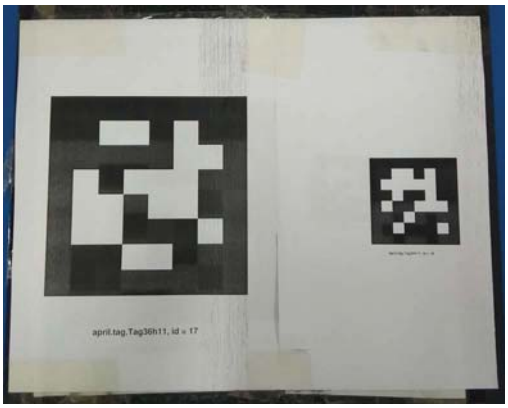


Fig.2: Landing target

In Fig. 3(a), the drone is at a higher altitude and a larger tag was detected. In 3(b), the drone is at a lower height, and two tags of different sizes were detected, and the data provided by the smaller tags were selected for more accurate positioning



(a) (b)

Fig. 3: The detection of AprilTags at different altitudes

3 System implementation

This section describes the hardware components of the system, the transformation of the coordinate system and the application of the PID control algorithm in this system.

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3.1 System hardware

The entire system includes a DJI M100 drone, a landing target of AprilTags, a router, a cell phone, and a ground unmanned vehicle.

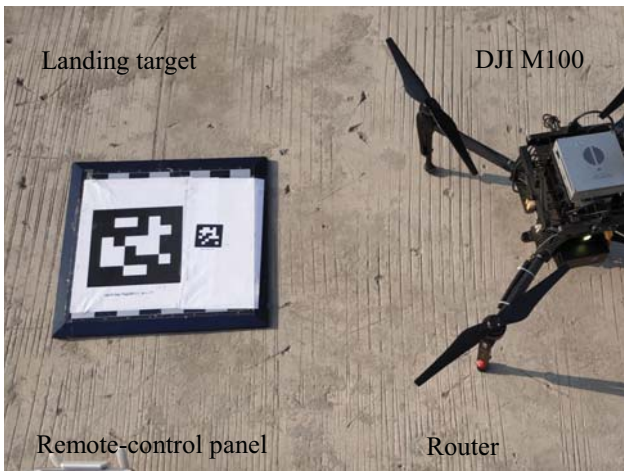


Fig. 4: System hardware

The M100 is equipped with a Manifold computer and a Microsoft HD3000 camera. The onboard computer guarantees real-time image processing. The router is used to connect the onboard computer and mobile phone through the wireless LAN. We send commands remotely through the mobile phone. The program on the drone can be started.

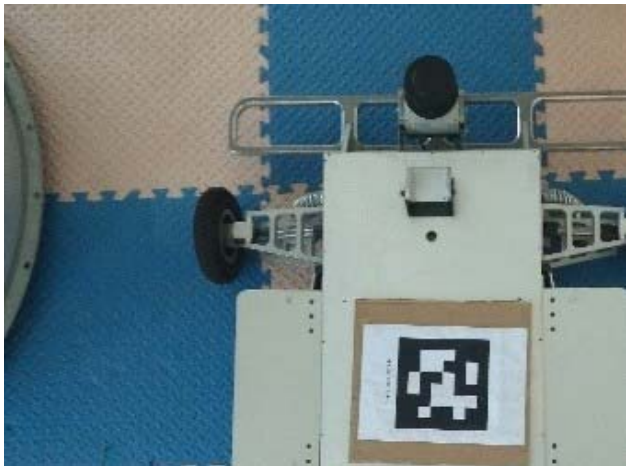


Fig. 5: Ground vehicle

The ground landing platform of the drone is equipped with an Intel NUC that serves as the main control computer. It is a ground mother ship platform designed for the UAV cooperation mechanism based on ROS. In order to make it have good software compatibility and practicability, its drive design adopts tank-type track differential drive mode and is equipped with a classic three-loop structure of speed, current, and position. The lower mechanical design is also equipped with a damping mechanism and protection mechanism, which can be adapted to a more complex environment. In order to ensure the landing area when cooperating with the aircraft, the upper mechanical design also has a foldable double wing device. The wireless controller controls the movement of the vehicle through the ROS publishing node message.

3.2 Coordinate transformation

The whole system involves a total of five coordinate systems' conversion: pixel coordinate system, image coordinate system, camera coordinate system, world coordinate system, and the body coordinate system.

The first is the conversion between the pixel coordinate system and the image coordinate system.

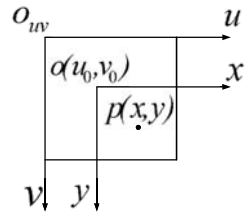


Fig. 6: Pixel coordinate system and image coordinate system

Both the pixel plane and the image plane are on the imaging plane, except that the respective origins and units of measurement are different. The origin of the image coordinate system is the intersection of the camera's optical axis and the imaging plane, usually the midpoint of the imaging plane. The unit of the image coordinate system is mm, which belongs to the physical unit, and the unit of the pixel coordinate system is pixel. We usually describe a pixel as a few rows and columns, so the relationship between the two is as follows:

$$u = \frac{x}{dx} + u_0 \tag{1}$$

$$v = \frac{y}{dy} + v_0 \tag{2}$$

u and v in equations (1) and (2) are the coordinates of the image point $p(x,y)$ in the pixel plane, u_0 and v_0 are the principal point offsets of the image coordinate system in the pixel coordinate system. The form of the matrix is written as:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{dx} & 0 & u_0 \\ 0 & \frac{1}{dy} & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \tag{3}$$

Then there is the conversion of the world coordinate system and the camera coordinate system.

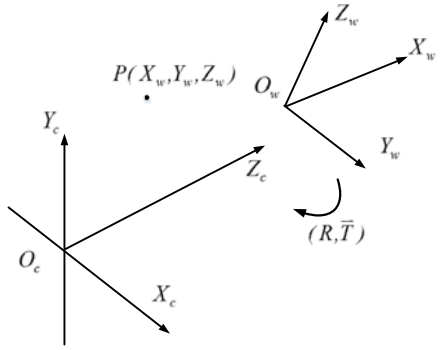


Fig.7: World coordinate system and camera coordinate system

In Fig. 7, $P(X_w,Y_w,Z_w)$ is a point in the world coordinate system, which is the true point in life. It can be

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converted from the world coordinate system to the camera coordinate system by a certain rotation and translation.

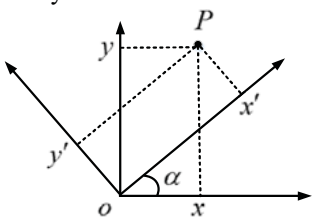


Fig.8: Rotation of the coordinate system

Fig.8 shows the rotation of the coordinate system around the X-axis. The coordinates of the point P under the coordinate system and the coordinates under the coordinate system satisfy the following relationship:

$$x' = x\cos\alpha - y\sin\alpha \tag{4}$$

$$y' = x\sin\alpha + y\sin\alpha \tag{5}$$

So, the rotation matrix around a single coordinate axis can be written as:

$$R_z(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix} \tag{6}$$

The same principle can be obtained by $R_x(\beta)$ and $R_y(\theta)$. The rotation matrix of the coordinate system rotating around x-axis and y-axis, according to:

$$R=R_z(\alpha)R_y(\theta)R_x(\beta) \tag{7}$$

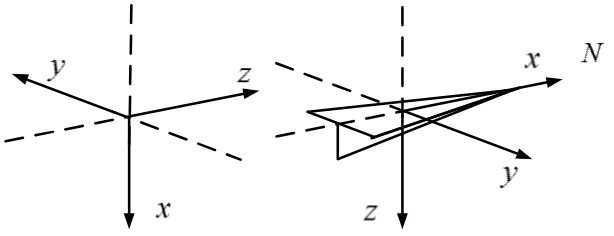
Finally, we get the rotation matrix R in three directions:

$$R=\begin{bmatrix} \sin\theta\cos\alpha & \sin\beta\sin\theta\cos\alpha - \cos\beta\sin\alpha & \cos\beta\sin\theta\cos\alpha + \sin\beta\sin\alpha \\ \cos\theta\sin\alpha & \sin\beta\sin\theta\cos\alpha + \cos\beta\cos\alpha & \cos\beta\sin\theta\sin\alpha - \sin\beta\cos\alpha \\ -\sin\theta & \sin\beta\sin\theta\cos\alpha + \cos\beta\cos\alpha & \cos\beta\cos\theta \end{bmatrix} \tag{8}$$

The translation matrix $\overline{T}$ in Fig,7 represents the amount of translation of the world coordinate system and the camera coordinate system in the x , y , and z directions.

According to the conversion relationship between the world coordinate system and the camera coordinate system, the relative position and attitude between the AprilTags and the camera can be obtained.

Finally, we need to figure out the relationship between the frame coordinate system and the camera coordinate system.



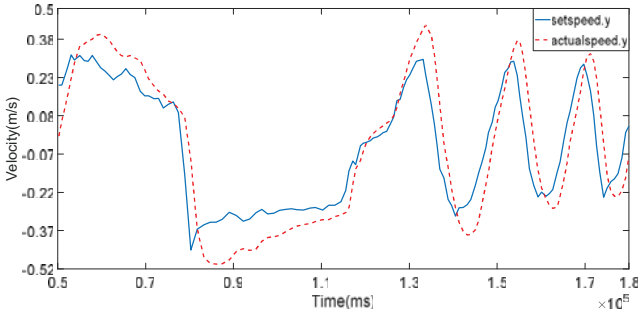
(a) (b)

Fig.9: camera coordinate system and body coordinate system

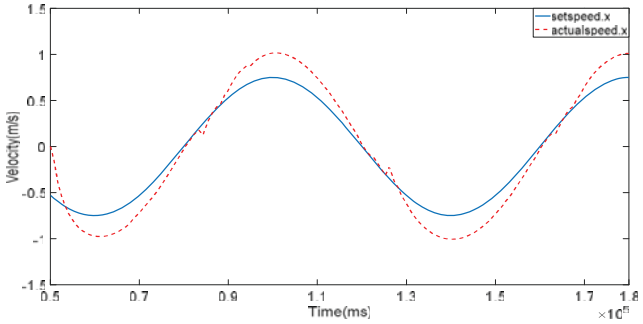
When the M100 UAV starts API control, its flight mode is headless, flying in the North East Down(NED) coordinate system, as shown by Fig.9(b). The camera coordinate system, as shown by Fig.9(a), takes the center of light as the origin, and the y-axis is opposite to the UAV coordinate system. After understanding the relationship between the two coordinate systems, the API function of the detected relative deviation to the UAV can be used to control the UAV.

3.3 PID control

Since when the API function of M100 is called directly, the system will have some overshoot and lag. As shown in Fig.10, the blue line represents the velocity input value of the UAV, and the red line represents the velocity response value. The velocity input of UAV in 10 (a) is the displacement deviation between the actual camera and the landing target. In 10 (b), the velocity input of UAV is a sinusoidal signal, and the PID control algorithm is based on its good dynamic stability. It is widely used in engineering [13], so we choose the classical PID control algorithm to solve the issue of response overshoot and lag. Figure 11 shows the basic structure of the PID controller.



(a)



(b)

Fig. 10: Velocity response without PID control

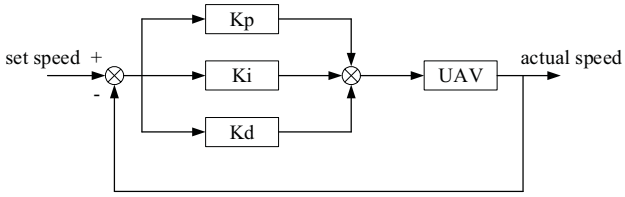


Fig.11 PID controller

Because the speed of the UAV is directly proportional to the position offset, the offset between the UAV and the tag is taken as the input speed of the UAV in a certain proportion, in order to solve this problem, the PID control law of speed control is given:

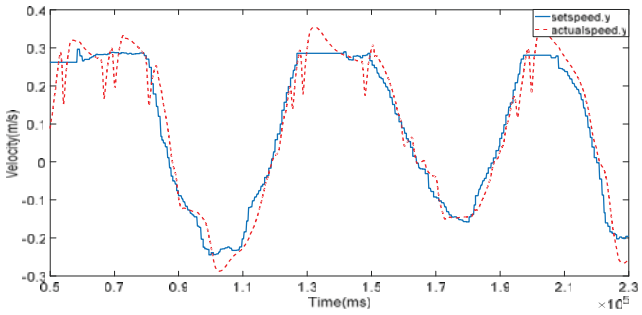
$$V=k_p \cdot err + k_i \cdot integral + k_d \cdot (err - last_err) \tag{9}$$

In the formula (9), k_p , k_i ,and k_d are the proportional, integral, and differential coefficients, respectively, and err is the error at the current time. The $last_err$ is the error of

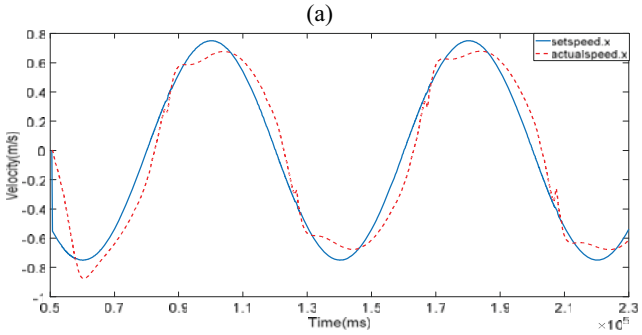
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the previous moment. In this paper, k_p=0.20 , k_i=0.03 , k_d=0.35 .

It can be seen from Fig.12 that the PID control algorithm can effectively solve the overshoot problem of the system and increase the dynamic stability of the system.



(a)



(b)

Fig.12: Velocity response with PID algorithm

4 Experiment

In this section, two experiments are designed: fixed-point landing experiment and dynamic tracking experiment to prove the feasibility and effectiveness of the proposed algorithm.

4.1 Landing experiment on a static platform

In this experiment, the drone takes off from a fixed position. After reaching a certain height, it starts to search for the ground landing target and then moves closer to the landing target. First, when the position difference between the drone and the ground landing target is less than a certain threshold, the landing begins. Landed on the ground platform.



Fig.13: landing experiment with a static target

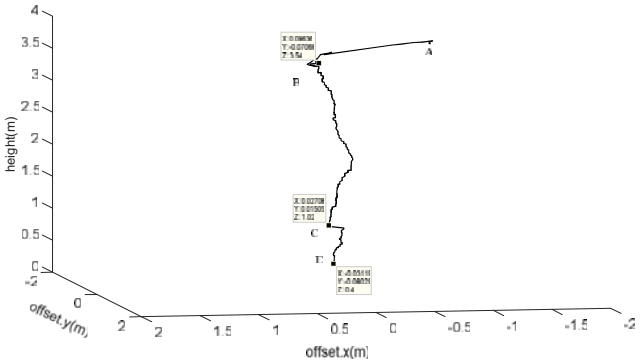


Fig.14: Landing trajectory

In Fig.14, the drone takes off to point A and then starts to move toward the label with a larger side length. It reaches the position of point B directly above the landing target and starts to land. After descending to point C, it changes to the identification side length. The small tag, after reaching D point close enough to the ground, starts the landing procedure and landed on the landing target. The key points' coordinate of the trajectory can be seen in Table 1. The error of the final landing position of the drone in the x and y directions is less than 1%.

Table 1: Key point of the trajectory

	B	C	D
X(m)	0.09636	0.02708	-0.03119
Y(m)	-0.07068	0.01505	-0.08029
Z(m)	3.54	1.02	0.4

4.2 Dynamic tracking experiment

In this experiment, the drone automatically takes off from the ground platform and starts to track the ground mobile platform after reaching a certain height. When the relative offset between the drone and the ground mobile platform is less than a certain value, the drone starts to land. Finally, Landed on the ground platform.



Fig.15: Dynamic tracking test

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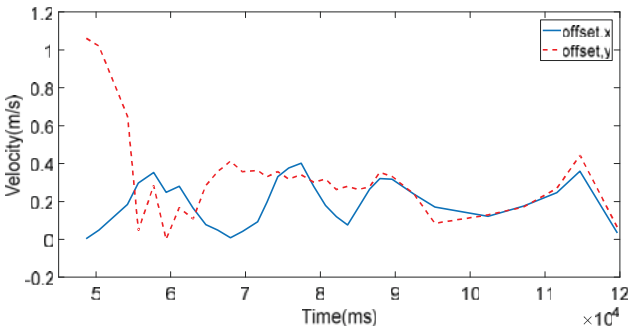


Fig.16: Position offset on X and Y axes

Fig.16 is the only offset between the X-axis and Y-axis directions of the UAV and the ground vehicle in the course of tracking the ground mobile platform. It can be seen that the offset between the two has remained within the range of (-0.2m, +0.5m).

5 Conclusion

In this paper, the motion control, target detection, system communication and other issues involved in the autonomous landing and tracking system of UAV are studied. Firstly, the principle of AprilTags visual positioning algorithm is expounded. According to the relationship between the size of AprilTags tag and positioning accuracy, two tags of different sizes are designed as landing targets on the landing platform. Then, according to DJI M100 UAV, two tags of different sizes are designed. In order to improve the dynamic tracking effect of UAV, the PID control algorithm is combined with the UAV control system. Finally, the fixed-point landing experiment results show that UAV has better positioning accuracy at different altitudes, and the positioning error of the final landing position is less than 1%. The dynamic tracking experiment results show that in the process of UAV tracking the ground platform, the position offset is always within the range of (-0.2m, +0.5m), which has a good tracking effect. This paper provides a useful reference for the autonomous landing and tracking of UAV.

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